

PROMPT ENGINEERING MASTERCLASS

Advanced Prompt Engineering & Output Control

From Vague Instructions to Reliable AI Outputs

Module Overview

Prompt engineering is the practice of designing clear, structured instructions for AI systems.

The quality of an AI response depends heavily on the quality of the instructions it receives.

When instructions are vague, outputs often become:

- Generic
- Inconsistent
- Incomplete
- Difficult to reuse
- More likely to contain incorrect assumptions

When instructions are precise and structured, outputs become:

- Clear
- Repeatable
- Controllable
- Consistent
- Easier to evaluate
- Easier to reuse across workflows

CORE PRINCIPLE

Better Instructions → Better Outputs

Output quality is a direct reflection of instruction quality.

2.0 — WHY PROMPT ENGINEERING MATTERS

AI responds to the instructions and context it is given.

It cannot reliably infer your exact objective, audience, business context, preferred format, or success criteria unless those requirements are clearly communicated.

Weak Prompt

Generate a report.

Why It Fails

- No clear purpose
- No domain
- No audience
- No structure
- No success criteria

Strong Prompt

You are a business analyst.

Create a monthly performance report for a SaaS startup.

Focus on:

- Revenue growth
- Churn rate
- Customer acquisition cost

Use bullet points and include a short executive summary at the beginning.

What Changed?

The stronger prompt defines:

Role → Task → Context → Focus → Format

2.1 — PROMPT STRUCTURE

The Foundation of Control

Structured prompts reduce ambiguity and make AI outputs more predictable.

A strong prompt should answer four fundamental questions:

1. **Who should the AI act as?**

2. **What exactly should it do?**
 3. **What rules or constraints should it follow?**
 4. **What should the final output look like?**
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2.1.1 — ROLE → INSTRUCTION → CONSTRAINTS → EXAMPLES

This is one of the simplest structures for improving a prompt.

1. Role

Define the perspective, expertise, or responsibility the AI should adopt.

Examples:

- Business Analyst
- Performance Marketer
- Technical Writer
- Product Manager
- Financial Analyst
- UX Researcher

2. Instruction

Clearly define the task.

Avoid vague instructions such as:

Analyze this.

Instead, write:

Analyze the campaign performance and identify the three largest causes of declining conversions.

3. Constraints

Define boundaries.

Examples:

- Maximum 500 words
- Use simple language
- Do not make assumptions

- Use only the supplied data
- Return exactly five recommendations
- Avoid unnecessary technical terminology

4. Examples

Show the AI what good output looks like whenever format, tone, or style is important.

Weak Prompt

Explain cloud computing.

Strong Prompt

You are a technical writer.

Explain cloud computing to non-technical startup founders.

Keep the explanation under 200 words, use simple language, and include one real-world analogy.

2.1.2 — PRECISION OUTPUT FORMATS

Good prompts do not only define **what the AI should produce**.

They also define **how the answer should be delivered**.

Unstructured responses are harder to reuse inside workflows, documents, applications, and automation systems.

Instructions define the task.

Formats define the container.

Common Output Formats

Format	Best Used For
Markdown	Documentation, blogs, internal knowledge bases
JSON	APIs, automation, structured data, dashboards

Format	Best Used For
XML	Enterprise integrations and structured systems
Tables	Comparisons, reports, structured analysis
Bullet Points	Summaries, recommendations, action items

Example

Instead of:

Analyze these competitors.

Use:

Analyze the competitors and return the result as a Markdown table with the following columns:

- Company
- Target Audience
- Core Offer
- Strength
- Weakness
- Opportunity

2.1.3 — CONTEXT LOADING & REFERENCE MATERIALS

AI cannot automatically understand your business, project, customer, product, or internal data unless relevant context is provided.

Context may include:

- Financial data
- Product specifications
- Brand guidelines
- Customer research
- Technical documentation
- Previous reports
- Campaign data
- Company policies
- Reference examples

Weak Prompt

Use previous financial data.

Strong Prompt

Use the following Q3 financial summary as the only data source for this analysis.

Context Loading Formula

Background + Relevant Data + Source Rules + Task

Example

Background:

We are a B2B SaaS company selling project-management software.

Data:

Use the Q3 customer acquisition report attached below.

Source Rule:

Do not introduce external statistics or assumptions.

Task:

Identify the three largest acquisition problems and recommend actions for Q4.

2.1.4 — SYSTEM PROMPTS VS USER PROMPTS

AI applications may contain different layers of instructions.

System Prompt

Defines persistent operating behavior such as:

- Role
- Tone
- Boundaries
- Safety rules
- Response behavior
- General operating principles

User Prompt

Defines the specific task being requested at that moment.

SIMPLE ANALOGY

System Prompt = Operating Rules

User Prompt = Task Execution

2.2 — PROMPTING FRAMEWORKS

Structured Thinking Models

Prompting frameworks convert vague intent into structured instructions.

There is no single framework that should be used for every situation.

The correct framework depends on:

- Complexity of the task
- Amount of available context
- Required output control
- Need for repeatability
- Number of variables involved

This module introduces four practical systems:

1. **CRISP**
 2. **IDEA**
 3. **RACE**
 4. **7-Step Prompting System**
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2.2.1 — CRISP FRAMEWORK

Context · Role · Instruction · Steps · Purpose

Best Used When

Use CRISP when a task involves:

- Multiple variables
- Several dependencies
- Analysis
- Decision-making
- Structured execution

CRISP Structure

C — Context

What does the AI need to understand before starting?

R — Role

Who should the AI act as?

I — Instruction

What exactly needs to be done?

S — Steps

What process should the AI follow?

P — Purpose

Why is the output required?

CRISP PROMPT TEMPLATE

You are a **{role}**.

Context:

{background information, relevant data, constraints}

Instruction:

{clear task to perform}

Steps:

1. {step one}
2. {step two}
3. {step three}

Purpose:

{why the output is needed and how it will be used}

CRISP EXAMPLE — FINANCE

You are a financial analyst.

Context:

A mid-sized manufacturing company wants to reduce operating costs without affecting production capacity.

Instruction:

Analyze the current cost structure and identify three areas with the highest potential for savings.

Steps:

1. Review fixed versus variable costs
2. Identify operational inefficiencies
3. Recommend cost-optimization actions

Purpose:

Support leadership decisions for the next fiscal quarter.

2.2.2 — IDEA FRAMEWORK

Instruction · Data · Examples · Adjustments

Best Used When

Use IDEA when you need to:

- Create content
- Rewrite existing material
- Work from supplied information
- Replicate a specific style
- Control tone or presentation

IDEA Structure

I — Instruction

What should be created or improved?

D — Data

What facts, inputs, or source material should be used?

E — Examples

What examples demonstrate the desired result?

A — Adjustments

What should be changed regarding tone, length, audience, format, or constraints?

IDEA PROMPT TEMPLATE

Instruction:

{what to create or improve}

Data:

{facts, inputs, references}

Examples:

{sample style, format, benchmark}

Adjustments:

{tone, length, audience, constraints}

IDEA EXAMPLE — CONTENT WRITING

Instruction:

Write a blog article about remote-work productivity.

Data:

Use the supplied research showing how increased flexibility affects employee retention.

Examples:

Use short paragraphs, practical tips, and a neutral writing style.

Adjustments:

Keep the article under 800 words and finish with actionable takeaways.

2.2.3 — RACE FRAMEWORK

Role · Action · Context · Expectation

Best Used When

Use RACE when you need:

- Fast prompting
- Professional outputs
- Simple structure
- Clear expectations
- Minimal prompt complexity

RACE Structure

R — Role

Who should the AI act as?

A — Action

What should it do?

C — Context

What situation or background does it need?

E — Expectation

What should the result look like?

RACE PROMPT TEMPLATE

Role:

{who the AI should act as}

Action:

{specific task}

Context:

{relevant background}

Expectation:

{format, tone, success criteria}

RACE EXAMPLE — HR

Role:

HR Operations Manager

Action:

Summarize the interview feedback for a software engineer candidate.

Context:

The candidate completed both technical and cultural interviews.

Expectation:

Return a concise summary containing:

- Strengths
 - Concerns
 - Final recommendation
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2.2.4 — THE 7-STEP PROMPTING SYSTEM

Maximum Output Control

Best Used When

Use the 7-Step System when:

- Output quality is critical
- Tasks will be repeated
- Multiple team members will use the prompt
- Results need to follow a consistent structure
- The prompt will be used in a production workflow

The 7 Components

1. Objective

What final result must be achieved?

2. Role

What expertise should the AI adopt?

3. Audience

Who will consume the output?

4. Format

How should the response be structured?

5. Process

What steps should the AI follow?

6. Constraints

What rules and limitations apply?

7. Evaluation

How should the final answer be judged?

7-STEP PROMPT TEMPLATE

Objective:

{desired end result}

Role:

{AI role}

Audience:

{who will consume the output}

Format:

{Markdown, JSON, bullets, report, table, etc.}

Process:

{steps to follow}

Constraints:

{rules, limits, exclusions}

Evaluation:

{criteria used to judge quality}

7-STEP EXAMPLE — TECHNOLOGY

Objective:

Create a troubleshooting guide for API timeouts.

Role:

Senior Backend Engineer

Audience:

Internal Engineering Team

Format:

Markdown documentation

Process:

1. Identify likely causes

2. Explain how to diagnose each cause
3. List possible fixes
4. Include practical examples

Constraints:

Avoid vendor-specific solutions.

Evaluation:

Judge the final response based on clarity, accuracy, and completeness.

2.2.5 — DOMAIN FRAMEWORKS INSIDE PROMPTS

Prompt frameworks can also contain specialized models or processes.

Instead of asking AI to approach a problem generically, instruct it to apply a specific framework.

GENERIC DOMAIN PROMPT TEMPLATE

You are a **{role}**.

Apply the **{framework name}** framework.

Context:

{domain-specific background}

Task:

{what needs to be produced}

Output Format:

{structured expectation}

EXAMPLE — FITNESS

You are a fitness program designer.

Apply a **Goal** → **Plan** → **Progression** framework.

Context:

The client wants to improve endurance over 12 weeks.

Task:

Design a phased training plan.

Output Format:

Provide a weekly breakdown with progression notes.

2.3 — PROMPT DEBUGGING, TESTING & VERIFICATION

Core Mindset

When an AI repeatedly produces inconsistent results, inspect the instructions.

Inconsistent Output = Signal to Inspect the Prompt

Use this improvement cycle:

Design → Test → Evaluate → Refine

2.3.1 — SELF-CHECK LOOPS

AI can be instructed to review its response against predefined criteria.

Self-Check Prompt

Review the response above for:

- Logical gaps
- Unclear assumptions
- Missing steps
- Contradictions
- Unsupported claims

Then improve the response where necessary.

2.3.2 — CRITIC—CREATOR FRAMEWORK

Instead of accepting the first response, use a controlled improvement loop.

Workflow

Step 1 — Create

Generate the first version.

Step 2 — Critique

Evaluate the output against predefined criteria.

Step 3 — Improve

Rewrite the output using the critique.

Simple Flow

Create → **Critique** → **Improve**

CRITIC–CREATOR TEMPLATE

Creator Phase:

Produce the requested output.

Critic Phase:

Review the output against:

- Accuracy
- Clarity
- Completeness
- Relevance
- Compliance with instructions

Improvement Phase:

Rewrite the output by correcting the issues identified during the critique.

2.3.3 — A/B TESTING PROMPTS

Prompt optimization should not rely entirely on intuition.

Create multiple versions of a prompt and compare their outputs.

Process

Prompt A → **Output A**

Prompt B → Output B

Then compare them using the same success criteria.

Common Use Cases

- Ad copy generation
- Product messaging
- UX microcopy
- Email writing
- Content hooks
- Reports

2.3.4 — 10/10 PROMPT QUALITY SCORECARD

Before using an important prompt repeatedly, evaluate it.

Parameter	Score
Role clearly defined	☐
Objective clearly defined	☐
Relevant context provided	☐
Constraints included	☐
Output format specified	☐
Audience clearly defined	☐
Examples provided where needed	☐
Verification step included	☐
Hallucination control included	☐
Prompt is reusable	☐

Scoring

0 = Missing

1 = Included

Maximum Score = 10/10

Score Interpretation

Score	Quality Level
0–4	Unreliable
5–7	Inconsistent
8–9	Strong
10	Production-Grade

2.3.5 — HALLUCINATION CONTROL SYSTEM

Hallucination risk increases when AI has to infer missing information.

Prompt design can reduce this risk.

Control Techniques

1. Limit Sources Explicitly

Use only the information contained in the attached report.

2. Require Assumption Disclosure

If you make an assumption, label it clearly as an assumption.

3. Add Verification Instructions

Verify every recommendation against the supplied data before finalizing the answer.

4. Allow Missing-Data Responses

If the available information is insufficient to answer accurately, state what information is missing instead of inventing an answer.

HALLUCINATION CONTROL TEMPLATE

Use only the supplied reference material.

Do not invent statistics, facts, names, or examples.

Clearly identify any assumption you make.

If required information is unavailable, state:

"Insufficient information provided."

Before finalizing the response, verify that every factual claim is supported by the supplied material.

2.4 — MULTI-STEP PROMPT DESIGN

Complex tasks should not always be handled using one large instruction.

Breaking a task into stages gives greater control over the workflow.

2.4.1 — STEP-BY-STEP TASK DECOMPOSITION

Complex tasks can be divided into smaller stages.

Useful for:

- Financial forecasting
- Technical troubleshooting
- Strategic planning
- Research
- Business analysis

Example

Instead of:

Create a complete market-entry strategy.

Break it into:

1. Analyze the market
 2. Identify customer segments
 3. Analyze competitors
 4. Evaluate positioning opportunities
 5. Recommend an entry strategy
 6. Review the recommendation against defined criteria
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2.4.2 — CHAIN OF PROMPTS

A Chain of Prompts uses several prompts sequentially.

Each prompt produces information required by the next stage.

Typical Sequence

Research → Structure → Draft → Review → Optimize

Example Workflow

Prompt 1 — Research

Analyze the supplied customer research and identify the five most important customer problems.

Prompt 2 — Structure

Organize those five problems into a logical landing-page messaging structure.

Prompt 3 — Draft

Write landing-page copy using the approved structure.

Prompt 4 — Review

Critique the copy for clarity, relevance, and persuasion.

Prompt 5 — Optimize

Rewrite the copy using the critique while maintaining the original positioning.

2.4.3 — PLANNER PROMPTS

Planner prompts separate planning from execution.

Prompt 1

Create the plan.

Prompt 2

Execute the plan.

Common Applications

- Software development
 - Business operations
 - Content pipelines
 - Research workflows
 - Complex reports
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2.4.4 — MULTI-MODEL COORDINATION

Different AI models can be assigned different responsibilities inside the same workflow.

Model 1 — Creation

Produces the initial output.

Model 2 — Validation

Reviews the output for errors or weaknesses.

Model 3 — Compression

Converts the validated output into a shorter final format.

Example Flow

Creation → Validation → Compression → Final Output

2.5 — PROMPT VERSION CONTROL & TEAM USAGE

Prompts used repeatedly should be treated as reusable business assets.

Important prompts should not exist as random messages scattered across individual chats.

Recommended Practices

1. Version Labels

Examples:

- Landing_Page_Copy_v1.0
- Landing_Page_Copy_v1.1
- Landing_Page_Copy_v2.0

2. Performance Notes

Record:

- What worked
- What failed
- What changed
- Why it changed

3. Shared Repositories

Store important prompts in one organized location.

4. Standard Templates

Create reusable structures so team members do not rebuild prompts from scratch.

PROMPT DOCUMENTATION FORMAT

Prompt Name:

{prompt name}

Version:

{version}

Purpose:

{what the prompt does}

Inputs Required:

{required information}

Expected Output:

{output structure}

Performance Notes:

{observations}

Last Updated:

{date}

2.6 — BUILDING A 40-PROMPT BUSINESS LIBRARY

A strong AI workflow becomes more valuable when effective prompts are converted into reusable assets.

Instead of writing a new prompt every time, build a structured library.

Core Prompt Categories

1. Research

- Competitive analysis
- Market research
- Customer research
- Data synthesis
- Trend analysis

2. Editorial

- Articles
- Video scripts
- Social content
- Content calendars
- Content repurposing

3. Operations

- Standard operating procedures
- Internal reports
- Meeting summaries
- Process documentation

4. Human Resources

- Job descriptions
- Interview summaries
- Candidate evaluations
- Employee communication

5. Sales

- Outreach emails
- Proposals
- Follow-up messages
- Sales research

6. Decision Support

- Trade-off analysis
- Recommendation reports
- Option comparisons
- Risk analysis

Every Library Prompt Should Be

Role-Based

Clearly define who the AI should act as.

Input-Driven

Specify what information must be supplied.

Output-Consistent

Define a predictable structure for the response.

2.7 — MULTIMODAL PROMPT VERIFICATION

Prompt engineering is not limited to text.

Image, video, and audio generation also require clear instructions and verification criteria.

2.7.1 — IMAGE PROMPT VERIFICATION

Before finalizing an image-generation prompt, check:

- Subject clarity
- Visual style
- Composition
- Lighting
- Mood
- Aspect ratio

Common Applications

- Brand creatives
- Product imagery
- Fitness visuals

- Social media graphics
- Marketing assets

IMAGE PROMPT CHECKLIST

Subject:

{who or what should appear}

Style:

{visual aesthetic}

Composition:

{placement and framing}

Lighting:

{lighting direction/style}

Mood:

{emotion or atmosphere}

Aspect Ratio:

{16:9, 1:1, 9:16, etc.}

2.7.2 — VIDEO PROMPT VERIFICATION

Video generation requires control across time.

Check:

- Scene sequence
- Timing
- Camera motion
- Transitions
- Final frame

Common Applications

- Advertisements
- Explainer videos
- Social media reels
- Product demonstrations

VIDEO PROMPT CHECKLIST

Scene 1:

{opening visual}

Scene 2:

{next action}

Scene 3:

{progression}

Camera Motion:

{pan, zoom, tracking, static, etc.}

Transitions:

{how scenes connect}

Timing:

{duration or pacing}

Final Frame:

{how the video should end}

2.7.3 — AUDIO PROMPT VERIFICATION

Audio prompting requires more than providing a script.

Check:

- Voice type
- Tone
- Pace
- Accent
- Emotion cues

Common Applications

- Voiceovers
- Podcasts
- Narration
- Guided sessions
- Training material

AUDIO PROMPT CHECKLIST

Voice:

{voice type}

Tone:

{professional, conversational, energetic, calm, etc.}

Pace:

{slow, moderate, fast}

Accent:

{preferred accent if required}

Emotion:

{confidence, excitement, empathy, authority, etc.}

MODULE 2 — MASTER SUMMARY

Prompt engineering is not simply about writing longer prompts.

It is about designing instructions that give AI enough information and structure to produce the desired result consistently.

Core System

CONTEXT → INSTRUCTION → CONTROL → VERIFICATION → REFINEMENT

What You Learned

By completing this module, you now understand how to:

- Structure prompts for greater control
- Assign clear AI roles
- Define precise tasks and constraints
- Control output formats
- Load reference material and context
- Distinguish system prompts from user prompts
- Use CRISP, IDEA, RACE, and the 7-Step Prompting System
- Embed domain frameworks inside prompts
- Debug inconsistent prompts
- Build self-check and critique loops
- A/B test prompt variations

- Evaluate prompts with a quality scorecard
 - Reduce hallucination risk
 - Break complex tasks into multiple stages
 - Build chains of prompts
 - Separate planning from execution
 - Coordinate different AI responsibilities
 - Version and document reusable prompts
 - Build a structured business prompt library
 - Verify image, video, and audio prompts
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THE PROMPT ENGINEERING CONTROL LOOP

1. DEFINE

What exactly do you want?



2. STRUCTURE

Convert the requirement into clear instructions.



3. CONTROL

Add context, constraints, format, and examples.



4. GENERATE

Run the prompt.



5. VERIFY

Check the response against success criteria.



6. REFINE

Improve the prompt based on the result.



7. REUSE

Save successful prompts as reusable assets.

FINAL PRINCIPLE

Do not treat prompts as one-time messages.

Treat important prompts as **designed systems that can be tested, improved, versioned, and reused.**

QUICK REFERENCE — WHICH FRAMEWORK SHOULD I USE?

Situation	Recommended Framework
Fast professional task	RACE
Content creation or refinement	IDEA
Complex task with multiple steps	CRISP
Highly controlled repeatable workflow	7-Step System
Improving an existing output	Critic–Creator
Comparing prompt performance	A/B Testing
Complex multi-stage project	Chain of Prompts
High-risk factual task	Hallucination Control System

MODULE 2 COMPLETE

Advanced Prompt Engineering & Output Control

Next Stage:

Applying structured prompt engineering to increasingly automated, coordinated, and scalable AI workflows.